

Zhaosen Qu

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RESEARCH INTERESTS

Ferroelectric and emerging non-volatile memory devices; nanoscale semiconductor device fabrication and scaling; thin-film deposition and process integration; wide-bandgap nitride synthesis and characterization for quantum photonic integrated circuits.

EDUCATION

Johns Hopkins University

B.S. in Electrical Engineering, Materials Science & Engineering
Minor in Mathematics
Dean's List: Fall 2024, Spring 2025, Fall 2025

Baltimore, MD
Expected: May 2028
GPA: 3.99/4.00

Selected Coursework: Introduction to VLSI, Advanced Digital Systems, Physical Chemistry II (Quantum Mechanics), Electronic Properties of Materials, Thermodynamics of Materials

RESEARCH EXPERIENCE

Center for Nanophase Materials Sciences, Oak Ridge National Laboratory

Oak Ridge, TN

DOE ORISE Research Student Intern (Incoming) | Mentor: Dr. Kyle Kelley

May – Aug 2026

- Will synthesize high-quality aluminum nitride (AlN) thin films on SiO₂/Si substrates and barium titanate (BTO) using controlled thin-film deposition techniques, optimizing growth parameters to minimize grain boundaries and defect centers for quantum photonic integrated circuits (QPICs).
- Will characterize film crystallinity and structural quality primarily through X-ray diffraction (XRD) analysis, including rocking curve measurements to achieve low-angle grain boundaries below 2 degrees for enhanced optical properties.
- Will investigate silicon oxynitride template layer formation to optimize AlN nucleation and interface quality, correlating growth conditions with crystallographic structure and photonic device performance metrics.

Weihs Group, Dept. of Materials Science and Engineering, Johns Hopkins University

Baltimore, MD

Undergraduate Research Assistant | PI: Prof. Timothy P. Weihs

Dec 2024 – Present

- Investigating structure–property relationships in Cu, CuTi, Al, and Ti alloys, focusing on how grain morphology, texture, and defect density govern electrical conductivity and mechanical hardness.
- Performing multi-modal characterization using SEM, EBSD, XRD, and Vickers microhardness testing to quantify grain size distributions, crystallographic texture, and dislocation density; elucidating strengthening mechanisms in PVD-deposited metallic foils.
- Preparing metallographic specimens via wire EDM, mechanical polishing, and controlled annealing to enable accurate and reproducible microstructural analysis; results contributed to a publication in *Materials Science and Engineering: A* (co-author).

Device Research and Engineering Laboratory, University of Pennsylvania

Philadelphia, PA

NSF REU Nanofabrication Research Intern | PI: Prof. Deep Jariwala

May – Aug 2025

- Scaled ferroelectric AlScN diode non-volatile memory devices from 200 μm to the 40 nm node, demonstrating viability for high-density (100 Mbit/mm²) storage; result to a publication in *Nano Letters* (co-author).
- Fabricated nanoscale devices in a Class 100 cleanroom using physical vapor deposition (PVD), electron-beam lithography, photolithography, and reactive ion etching (RIE); developed and optimized lift-off processes to improve yield and reproducibility at sub-50 nm feature sizes.
- Characterized ferroelectric switching behavior in sub-50 nm AlScN diodes via P–E hysteresis and I–V measurements, achieving a 20% reduction in coercive field and a 10 $\times$ decrease in leakage current through ITO/Al top contact engineering.
- Investigated proximity ferroelectricity in 5 nm AlN / 15 nm AlScN thin-film stacks, exploring interfacial coupling mechanisms to enhance polarization retention and device reliability.
- Correlated nanoscale surface morphology (AFM) with electrical performance metrics to systematically optimize sputter deposition parameters, directly informing process-of-record improvements.

PUBLICATIONS

3. R. Berlia, **Z. Qu**, Y. Sun, R. T. Mehta, and T. P. Weihs, "Combinatorial mapping of composition–processing–property relationships in precipitation-strengthened Cu–Ti alloys," *Materials & Design*, submitted for review, 2025.
2. R. Berlia, C. Singh, **Z. Qu**, R. T. Mehta, and T. P. Weihs, "Revealing bulk-like properties in freestanding PVD copper foils," *Materials Science and Engineering: A*, vol. 947, p. 149163, 2025. doi: 10.1016/j.msea.2025.149163.
1. Z. Hu, H. Cho, R. K. Rai, K. Bao, Y. Zhang, **Z. Qu**, Y. He, Y. Ji, C. Leblanc, K.-H. Kim, Z. Han, Z. Qiu, X. Du, E. A. Stach, R. Olsson, and D. Jariwala, "Demonstration of highly scaled AlScN ferroelectric diode memory with a storage density of 100 Mbit/mm²," *Nano Letters*, vol. 25, no. 37, pp. 13748–13755, 2025. doi: 10.1021/acs.nanolett.5c02961.

CONFERENCE PRESENTATIONS

2. **Z. Qu**, "Proximity Ferroelectricity in Thin Nitride-Ferroelectric Stack Using 5 nm AlN – 15 nm AlScN," poster presented at the *Singh REU Symposium*, Singh Center for Nanotechnology, University of Pennsylvania, Philadelphia, PA, July 31, 2025. PI: Prof. Deep Jariwala; Mentors: Zekun Hu, Dr. Yuhong Cao.
1. **Z. Qu**, Z. Hu, Y. Cao, and D. Jariwala, "Overcoming the Memory Wall: Scaling AlScN FeDiodes to sub-50 nm and Enhancing Reliability via Proximity Ferroelectricity," poster presented at the *Yale Undergraduate Research Conference*, New Haven, CT, February 2, 2026.

ADDITIONAL ENGINEERING EXPERIENCE

Space Radiation Lab, California Institute of Technology

Pasadena, CA

Electrical Engineer Intern | SPHEREx Space Telescope

Jun 2023 – Jul 2023

- Diagnosed and repaired overstressed components in the SPHEREx flight assembly by performing root-cause analysis and schematic review in Altium Designer, restoring system reliability for NASA mission qualification.
- Conducted accelerated stress-testing on circuit components using programmable power supplies and oscilloscopes to characterize failure modes; designed optical calibration fixtures in SOLIDWORKS.
- Managed Bill of Materials (BOM) revisions across multiple flight board versions, ensuring traceability and configuration control for mission-critical hardware.

TEACHING EXPERIENCE

Johns Hopkins University

Baltimore, MD

Undergraduate Teaching Assistant

Jan 2025 – Present

- Spring 2026: **Linear Algebra** (Teaching Assistant), **Intermediate Programming** (Course Assistant)
- Fall 2025: **Linear Algebra** (Teaching Assistant), **Intermediate Programming** (Course Assistant)
- Spring 2025: **Calculus II for Engineers** (Course Assistant), **Physics II for Engineers** (Learning Assistant)

TECHNICAL SKILLS

Nanofabrication: Photolithography, Electron-Beam Lithography, Reactive Ion Etching (RIE), Physical Vapor Deposition (Sputtering, E-beam Evaporation), Chemical Vapor Deposition (CVD), Lift-off Processing, Class 100 Cleanroom Operation

Materials Characterization: Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM), Electron Backscatter Diffraction (EBSD), Powder X-ray Diffraction (XRD), Vickers Microhardness Testing, Ferroelectric P–E Hysteresis and I–V Measurement

Electronics & EDA: Altium Designer, Cadence Virtuoso, Xilinx Vivado, LTspice, SystemVerilog, FPGA Design, SOLIDWORKS

Programming & Computation: Python, C, C++, MATLAB, Java

HONORS & AWARDS

NSF Research Experiences for Undergraduates (REU) Fellow, University of Pennsylvania
Dean's List, Johns Hopkins University (3 consecutive semesters)

2025
2024 – 2025